

scVI Human Validation — Reviewer 2 (Blinded)

Purpose

Independently determine how each citing paper uses scVI. Machine labels, confidence, agent outputs, and machine-selected evidence are intentionally hidden.

Workflow

Calibration: complete CAL-01 to CAL-03 independently, then discuss those three cases and freeze the codebook.

Formal validation: complete every HV row in the displayed order. Do not open the other reviewer document or the consensus document until both formal reviews are locked.

Reconciliation: after both reviewers lock their answers, the coordinator copies both records into the consensus document. Discuss only disagreements. Use `REFERENCE_UNRESOLVED` when agreement is not possible.

Evidence boundary: use only the linked approved article versions. Search all occurrences of “scVI” and inspect the surrounding methods, results, figures, tables, and supplements that are linked.

Allowed values

Citation validity: Yes / No / Unclear

Method executed: Yes / No / Unclear

Use labels (select all that apply): Benchmark / Extend or develop / Apply to biological data / Other

Primary use: Mention only / Benchmark / Extend or develop / Apply to biological data / Other / Unclear

Biological insight: Yes / No / Unclear

Evidence sufficiency: Sufficient / Incomplete / Citation not located

Supporting evidence: exact quote plus section, page, paragraph, figure, or table locator.

Review table

Article and approved evidence links	Reviewer 2 independent record
CAL-01 CALIBRATION Cell Subtype Classification via Representation Learning Based on a Denoising Autoencoder for Single-Cell RNA Sequencing 2021 IEEE Access DOI: 10.1109/access.2021.3052923 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Use labels (select all that apply): Mention only Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: To perform fundamental analysis for single-cell transcriptome dataset, the single-cell variational inference (scVI) model was introduced for probabilistic representation [34] and ACTINN implemented a DNN model to assign each cell a cell type Notes: Reviewer initials: Review completed:

CAL-02 CALIBRATION Eleven grand challenges in single-cell data science 2020 Genome biology DOI: 10.1186/s13059-020-1926-6 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Use labels (select all that apply): Mention only Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Some of them are expressly aimed at imputation (e.g., AutoImpute, DeepImpute, EnImpute, DCA, and scVI). But even if imputation is not the main focus, such methods can generate “imputed” expression values as an upshot of a model primarily focused on other tasks, like learning latent spaces, clustering, batch correction, or visualization (and often several of these tasks simultaneously). Notes: Reviewer initials: Review completed:
CAL-03 CALIBRATION Mosaic integration of spatial multi-omics with SpaMosaic 2024 DOI: 10.1101/2024.10.02.616189 Article page Full text 1 Supplement 1 Supplement 2 Supplement 3 Supplement 4 Supplement 5 Supplement 6	Review status: Citation validity: Yes Method executed: Yes Use labels (select all that apply): Benchmark / Apply to biological data Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Additionally, the cortical regions were assigned to different clusters by CellCharter (RNA) in section 4 compared to the other three sections, which is consistent with the UMAP visualizations, despite the inputs having been batch corrected with scVI⁴⁹ Notes: Reviewer initials: Review completed:

Article and approved evidence links	Reviewer 2 independent record
HV-008 FORMAL VALIDATION LAIOR: a hyperbolic neural ODE variational framework for interpretable single-cell manifold learning and trajectory inference 2026 Frontiers in Genetics DOI: 10.3389/fgene.2026.1838613 Article page Full text 1 Supplement 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We benchmarked LAIOR on 118 datasets (53 scRNA-seq and 65 scATAC-seq) spanning human and mouse tissues, cancer, development, and immune responses (Figure 1B). The comparison includes 23 baseline methods covering contrastive learning (scGCC, CLEAR), hyperbolic manifolds (Poincaré maps, PGM, LPGM, HWN, scDHMAP), modality-specific approaches (PeakVI, PoissonVI), trajectory inference (scTour), factor-based models (β-VAE, InfoVAE, TC-VAE, DIP-VAE), and generative frameworks (scVI, scGNN, CellBlast, SCALEX, scDAC, scDeepCluster, scDiffusion, siVAE). Notes: Reviewer initials: Review completed:

HV-047 FORMAL VALIDATION Deep learning in single-cell and spatial transcriptomics data analysis: advances and challenges from a data science perspective 2025 Briefings in Bioinformatics DOI: 10.1093/bib/bbaf136 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Use labels (select all that apply): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: For modeling uncertainty, scVI incorporates stochastic factors inherent in the sequencing process, providing a framework to better capture and account for variability in the data. Notes: Reviewer initials: Review completed:
HV-020 FORMAL VALIDATION SCIntRuler: guiding the integration of multiple single-cell RNA-seq datasets with a novel statistical metric 2024 Bioinformatics DOI: 10.1093/bioinformatics/btae537 Article page Full text 1	Review status: Locked Citation validity: Yes Method executed: No Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We acknowledge that the primary focus of this work is to illustrate SCIntRuler and not to comprehensively evaluate existing data integration methods. The review works by Argelaguet et al. (2021), Luecken et al. (2022), and Ryu et al. (2023) systematically compared more scRNA-seq data integration methods, such as scGen (Lotfollahi et al. 2019), LIGER (Liu et al. 2020), FastMNN (Zhang et al. 2019), and scVI (Lopez et al. 2018). Notes: Reviewer initials: Review completed:

Article and approved evidence links	Reviewer 2 independent record
HV-016 FORMAL VALIDATION AtlasLens: Metadata-centric exploration and analysis of single-cell atlases 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.07.10.735823 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Use labels (select all that apply): Primary use: Extend or Develop Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: AtlasLens works with both single-cell dataset and atlases containing rich, diverse metadata built using integration tools such as Seurat and scVI Notes: Reviewer initials: Review completed:
HV-040 FORMAL VALIDATION Best practices for single-cell analysis across modalities 2023 Nature Reviews Genetics DOI: 10.1038/s41576-023-00586-w Article page Full text 1	Review status: Locked Citation validity: Yes Method executed: No Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Depending on the complexity of the integration tasks, such as atlas integration, deep-learning approaches such as scANVI⁴⁰, scVI⁴¹ and scGen⁴² as well as linear-embedding models such as Scanorama⁴³ performed best, whereas for less complex integration tasks, Harmony³⁷ is the preferred method³⁵. Notes: Reviewer initials: Review completed:

HV-034 FORMAL VALIDATION Single-Cell Multi-Omics: Insights into Therapeutic Innovations to Advance Treatment in Cancer 2025 International Journal of Molecular Sciences DOI: 10.3390/ijms26062447 Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: No Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Summary of multi-omics and computational integration: Multi-omics methods enable joint profiling of molecular layers, such as transcriptome-genome, transcriptome-epigenome, and proteome-transcriptome. Computational tools, including Seurat, Harmony, LIGER, scVI, MOFA+, totalVI, MultiVI, UNIMF, and MultiMap, integrate these datasets by harmonising data and aligning modalities, aiding the interpretation of complex single-cell multi-omics data. Notes: Reviewer initials: Review completed:
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Article and approved evidence links	Reviewer 2 independent record
HV-037 FORMAL VALIDATION Assessing Markovian and Delay Models for Single-Nucleus RNA Sequencing 2023 Bulletin of Mathematical Biology DOI: 10.1007/s11538-023-01213-9 Article page Full text 1 Supplement 1 Supplement 2	Review status: NOT STARTED Citation validity: Yes Method executed: No Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: not sufficient Supporting quote + locator: Notes: Reviewer initials: Review completed:
HV-027 FORMAL VALIDATION Multi-modal Diffusion Model with Dual-Cross-Attention for Multi-Omics Data Generation and Translation 2025 DOI: 10.1101/2025.02.27.640020 Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: Yes Use labels (select all that apply): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: For comparison, we also generated multi-omics data using other simulators, including MultiVI¹⁶, CFGen¹⁷, and scDesign3¹⁸, as well as two single-modal generators: scVI¹⁹ and peakVI²⁰, which generate scRNA-seq and scATAC-seq data, respectively. Notes: Reviewer initials: Review completed:

HV-001 FORMAL VALIDATION Computational tools for analyzing single-cell data in pluripotent cell differentiation studies 2021 Cell Reports Methods DOI: 10.1016/j.crmeth.2021.100087 Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: No Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: others utilize neural network models to embed both datasets in a shared reduced dimension (Johansen and Quon, 2019; Lopez et al., 2018). Notes: Reviewer initials: Review completed:
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Article and approved evidence links	Reviewer 2 independent record
HV-026 FORMAL VALIDATION Tumor-infiltrating CD27 - IgD - regulatory B cells suppress cytotoxic CD8 + T cell responses in renal cell carcinoma 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.07.08.663720 Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: Yes Use labels (select all that apply): Primary use: Apply to biological data Biological insight: Yes Evidence sufficiency: Yes Supporting quote + locator: The atlases were merged into a single AnnData object and processed in scVI⁷¹ using the scanpy python package Notes: Reviewer initials: Review completed:
HV-042 FORMAL VALIDATION Pertpy: an end-to-end framework for perturbation analysis 2025 Nature Methods DOI: 10.1038/s41592-025-02909-7 Article page Full text 1 Supplement 1 Supplement 2 Supplement 3 Supplement 4	Review status: NOT STARTED Citation validity: Yes Method executed: Yes Use labels (select all that apply): Primary use: Extend or develop Biological insight: No Evidence sufficiency: Supporting quote + locator: Pertpy is implemented in Python and builds upon several scientific open-source libraries, including NumPy⁷⁰, Scipy⁷¹, JAX¹⁵, scikit-learn⁷², Pandas^{72,73}, AnnData²², scanpy³⁴, muon²³, NumPyro⁷⁴, OTT-JAX⁷⁵, blitzGSEA⁶⁹, PyTorch⁷⁶ and scvi-tools¹³ for omics data handling and matplotlib⁷⁷ and seaborn⁷⁸ for data visualization. Notes: Reviewer initials: Review completed:

HV-038 | FORMAL VALIDATION

Deep-learning methods for unveiling large-scale single-cell transcriptomes

2024 | Cancer Biology and Medicine

DOI: 10.20892/j.issn.2095-3941.2023.0436

[Article page](#)

[Full text 1](#)

Review status: NOT STARTED

Citation validity: Yes

Method executed: No

Use labels (select all that apply):

Primary use: Mention only

Biological insight: No

Evidence sufficiency: Yes

Supporting quote + locator:

Single-cell variational inference (scVI)⁵ takes batch information into a VAE for batch correction. Single-cell annotation using variational inference

(scANVI)¹⁶ is a semi-supervised adaptation of scVI.

Notes:

Reviewer initials:

Review completed:

Article and approved evidence links	Reviewer 2 independent record
HV-010 FORMAL VALIDATION Single-cell characterization of anti-LAG-3 and anti-PD-1 combination treatment in patients with melanoma 2023 Journal of Clinical Investigation DOI: 10.1172/jci164809 Article page Full text 1	Review status: Citation validity: Yes Method executed: Yes Use labels (select all that apply): Primary use: Apply to biology data Biological insight: Yes Evidence sufficiency: Yes Supporting quote + locator: To overcome the batch effect, we used a probabilistic framework to account for different nuisance factors of variation in an unsupervised manner with a deep generative modeling method scVI. Briefly, the transcriptome of each cell is encoded through a nonlinear transformation into a low-dimensional, batch-corrected, latent embedding. Notes: Reviewer initials: Review completed:
HV-018 FORMAL VALIDATION Hierarchical classification of immune cell transcriptomes at population-scale 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.05.30.728980 Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: Yes Use labels (select all that apply): Primary use: Apply to biology data Biological insight: Yes Evidence sufficiency: Yes Supporting quote + locator: To estimate the loss of biological information by common integration approaches, we tested integration with scVI on our Suco resource. Notes: Reviewer initials: Review completed:

HV-007 FORMAL VALIDATION Biological network-inspired interpretable variational autoencoder 2020 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2020.12.17.423310 Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: No Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: More recently, deep generative models such as Variational Autoencoders⁷ (VAEs) have proven to be extremely useful for the probabilistic modeling of single-cell transcriptomes, such as scVI or scGen 8–10 . Notes: Reviewer initials: Review completed:
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Article and approved evidence links	Reviewer 2 independent record
HV-036 FORMAL VALIDATION SECANT: a biology-guided semi-supervised method for clustering, classification, and annotation of single-cell multi-omics 2022 PNAS Nexus DOI: 10.1093/pnasnexus/pgac165 Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: Yes Use labels (select all that apply): Primary use: Extend and develop Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: We first assessed the performance of SECANT with simulation studies. For convenience, we simulated data from mixture multivariate Gaussian distribution to mimic the latent space of RNA data through scVI, where the distribution parameters including cluster-specific mean vector, covariance matrix, and cluster weight are obtained from fitting SECANT on a public PBMC dataset processed by scVI. Notes: Reviewer initials: Review completed:
HV-023 FORMAL VALIDATION Transcriptomic advances in studies of muscle stem cell aging: From bulk to single-cell and beyond 2026 Cell Research DOI: 10.1038/s41422-026-01240-w Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: No Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: High-dimensional, multi-modal muscle datasets serve as the central input for four key analytical pillars designed to resolve biological complexity. Pillar 1 (Denoising): removes technical noise to reveal true biological signals (e.g., using scVI). Notes: Reviewer initials: Review completed:

HV-043 | FORMAL VALIDATION

Benchmarking atlas-level data integration in single-cell genomics

2021 | Nature Methods

DOI: 10.1038/s41592-021-01336-8

[Article page](#)

[Full text 1](#)

[Full text 2](#)

Review status: NOT STARTED

Citation validity: Yes

Method executed: Yes

Use labels (select all that apply):

Primary use: Benchmark

Biological insight: No

Evidence sufficiency: Yes

Supporting quote + locator:

We selected 12 single-cell data integration tools: mutual nearest neighbors (MNN)¹² and its extension FastMNN¹², Seurat v3 (CCA and RPCA)¹³, scVI¹⁴ and its extension to an annotation framework (scANVI¹⁵), Scanorama¹⁶, batch-balanced k nearest neighbors (BBKNN)¹⁷, LIGER¹⁸, Conos¹⁹, SAUCIE²⁰ and Harmony²¹, one bulk data integration tool (ComBat²²)

Notes:

Reviewer initials:

Review completed:

Article and approved evidence links	Reviewer 2 independent record
HV-025 FORMAL VALIDATION PI3Ky in B cells promotes antibody responses and generation of antibody-secreting cells 2024 Nature Immunology DOI: 10.1038/s41590-024-01890-1 Article page Full text 1 Supplement 1 Supplement 2 Supplement 3 Supplement 4 Supplement 5 Supplement 6 Supplement 7 Supplement 8 Supplement 9 Supplement 10	Review status: NOT STARTED Citation validity: Yes Method executed: Yes Use labels (select all that apply): Primary use: Apply to Biological Data Biological insight: Yes Evidence sufficiency: Yes Supporting quote + locator: Thereafter, the filtered data were integrated and batch effects corrected using single-cell variational inference (scVI)⁵⁹. The scVI's latent representation was used to compute the neighborhood graph for Louvain clustering and Uniform Manifold Approximation and Projection (UMAP) visualization. Notes: Reviewer initials: Review completed:
HV-013 FORMAL VALIDATION COVID-Datathon: Biomarker identification for COVID-19 severity based on BALF scRNA-seq data 2021 arXiv (Cornell University) DOI: 10.48550/arxiv.2110.04986 Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: No Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: As a result, it may not be beneficial to apply typical scRNA-seq tools for biomarker identification and cell clustering on this normalized data set, such as scVI [Lopez et al., 2018], DESeq2 [Love et al., 2014] and SimCD [Niyakan et al., 2021], in which Negative Binomial (NB) distribution models and their extensions were developed to model scRNA-seq data. Notes: Reviewer initials: Review completed:

HV-022 | FORMAL VALIDATION

Transforming microfluidics for single-cell analysis with robotics and artificial intelligence

2025 | Lab on a Chip

DOI: 10.1039/d5lc00216h

[Article page](#)

[Full text 1](#)

Review status: NOT STARTED

Citation validity: Yes

Method executed: No

Use labels (select all that apply):

Primary use: Mention only

Biological insight: No

Evidence sufficiency: Yes

Supporting quote + locator:

For example, adaptive sampling algorithms can guide iterative single-cell RNA sequencing to focus on rare or transitional cell states. In the realm of data integration, cutting-edge AI models such as variational autoencoders (e.g., scVI¹⁶³ and TotalVI¹⁶⁴)

Notes:

Reviewer initials:

Review completed:

Article and approved evidence links	Reviewer 2 independent record
HV-033 FORMAL VALIDATION HoloCell: A Generative Foundation Model for Holistic Cellular Modeling 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.06.07.730684 Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: Yes Use labels (select all that apply): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: omatin accessibility and gene expression profiles from 817,740 fetal cells across 12 organs. For transcriptomic representation learning, HoloCell was compared with scGPT and scFoundation in the zero-shot setting, as well as scVI, which was fitted on the HDMA dataset. Notes: Reviewer initials: Review completed:
HV-049 FORMAL VALIDATION Trends and Potential of Machine Learning and Deep Learning in Drug Study at Single-Cell Level 2023 Research DOI: 10.34133/research.0050 Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: No Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: scvis [109], Notes: Reviewer initials: Review completed:

HV-029 | FORMAL VALIDATION

Relevant Applications of Generative Adversarial Networks in Drug Design and Discovery: Molecular De Novo Design, Dimensionality Reduction, and De Novo Peptide and Protein Design

2020 | Molecules

DOI: 10.3390/molecules25143250

[Article page](#)

[Full text 1](#)

Review status: NOT STARTED

Citation validity: Yes

Method executed: No

Use labels (select all that apply):

Primary use: Mention only

Biological insight: No

Evidence sufficiency: Yes

Supporting quote + locator:

From the experiments, it was indicated that the DR-A model had better performance than other widely used dimensionality reduction methods, including the PCA [80], Zero-Inflated Factor Analysis (ZIFA) [83], Single-cell Variational Inference (scVI) [82], Sparse Autoencoder for Unsupervised Clustering, Imputation, and Embedding (SAUCIE) [84], T-distributed stochastic neighbor embedding (t-SNE) [85], and Uniform Manifold Approximation and Projection (UMAP) [86]. It should be noted that the scVI and SAUCIE methods also employ deep learning approaches such as deep autoencoder algorithms, but not the GAN architecture. Moreover, the scVI method is based on the variational autoencoder structure [14] and conditional ZINB distributions [87].

Notes:

Reviewer initials:

Review completed:

Article and approved evidence links	Reviewer 2 independent record
HV-028 FORMAL VALIDATION Deep learning applications in single-cell genomics and transcriptomics data analysis 2023 Biomedicine & Pharmacotherapy DOI: 10.1016/j.biopha.2023.115077 Article page Full text 1	Review status: Citation validity: Yes Method executed: No Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Single-cell variational inference (scVI) is another DNN algorithm introduced by Lopez et al. [40]. ScVI is based on a hierarchical Bayesian model and uses a DNN to define the conditional probabilities, assuming either a negative binomial or a zero-inflated negative binomial distribution [40]. Lopez et al. show that scVI can accurately recover gene expression signals and impute the zero-valued entries, potentially enhancing the downstream analyses without adding any artifacts or false signals. Notes: Reviewer initials: Review completed:

HV-041 FORMAL VALIDATION Decoding disease and therapy through multiomics integration and systems analysis 2026 Briefings in Bioinformatics DOI: 10.1093/bib/bbag311 Article page Full text 1	Review status: Citation validity: Yes Method executed: No Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: yes Supporting quote + locator: It has demonstrated good performance in aligning scRNA-seq datasets from various patients and conditions without the overcorrection of biologically interesting signals. Similarly, scVI (single-cell variational inference) employs deep generative modelling with variational autoencoders to represent gene expression distributions, allowing for probabilistic batch correction and imputation [36]. Notes: Reviewer initials: Review completed:
HV-014 FORMAL VALIDATION Kinetic Foundation of the Zero-Inflated Negative Binomial Model for Single-Cell RNA Sequencing Data 2020 SIAM Journal on Applied Mathematics DOI: 10.1137/19m1253198 Article page Full text 1 Full text 2	Review status: NOT STARTED Citation validity: Yes Method executed: No Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Among these statistical models, the zero-inflated negative binomial (ZINB) model is the most widely used [20–31], where the zero-inflated part describes dropouts and the negative binomial part accounts for over-dispersion. Some other commonly used models are listed in Sec. 7. Notes: Reviewer initials: Review completed:

Article and approved evidence links	Reviewer 2 independent record
HV-015 FORMAL VALIDATION Recovering Single-cell Heterogeneity Through Information-based Dimensionality Reduction 2021 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2021.01.19.427303 Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: Yes Use labels (select all that apply): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: For comparison, we also tested PCA, ICA, scVI, and six rare cell type discovery tools: RaceID [8], GiniClust [9], CellSIUS [10], FIRE [11], GeoSketch [12], and Hopper [13]. Notes: Reviewer initials: Review completed:
HV-002 FORMAL VALIDATION Multimodal Hypersampling of the adult human heart resolves tissue-specific cell states and disease-associated variants 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.06.16.732617 Article page Full text 1	Review status: Citation validity: Yes Method executed: Yes Use labels (select all that apply): Primary use: Apply to biology data Biological insight: Yes Evidence sufficiency: Yes Supporting quote + locator: To confirm annotation robustness, we assessed label recoverability in an independent scVI⁷ latent space using cross-validated <i>k</i>-nearest neighbours classification, achieving a median accuracy of 83.5%, confirming that assignments are robust to the choice of integration framework (Supplementary Fig. S1c,d,e). Notes: Reviewer initials: Review completed:

HV-009 FORMAL VALIDATION Coding agents author interpretable single-cell embedding models from the literature 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.07.07.737048 Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: Yes Use labels (select all that apply): Primary use: Benchmark Biological insight: No Evidence sufficiency: No Supporting quote + locator: UCell and expiMap form one tight cluster high on the batch axis, well above PCA, scVI-census, UCE and scVI without its batch key. Only scVI lies further to the top-right (Figure 2a). Notes: Reviewer initials: Review completed:
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Article and approved evidence links	Reviewer 2 independent record
HV-006 FORMAL VALIDATION Implementing distinct spatial proteogenomic technologies: opportunities, challenges, and key considerations 2024 Clinical & Experimental Immunology DOI: 10.1093/cei/uxae077 Article page Full text 1	Review status: Citation validity: Yes Method executed: No Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: No Supporting quote + locator: Some widely used examples are Harmony [68], which excels with smaller datasets and scVI [69], which is more appropriate for complex integration tasks [70, 71]. Notes: Reviewer initials: Review completed:
HV-039 FORMAL VALIDATION SCOPE: Localizing fate-decision states and their regulatory drivers in single-cell differentiation 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.04.07.717037 Article page Full text 1	Review status: NOT STARTED Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes: Reviewer initials: Review completed:

HV-012 | FORMAL VALIDATION

sCellST: a Multiple Instance Learning approach to predict single-cell gene expression from H&E images using spatial transcriptomics

2024 | bioRxiv (Cold Spring Harbor Laboratory)

DOI: 10.1101/2024.11.07.622225

[Article page](#)

[Full text 1](#)

Review status: NOT STARTED

Citation validity: Yes

Method executed: Yes

Use labels (select all that apply):

Primary use: Extend or develop

Biological insight: No

Evidence sufficiency: Yes

Supporting quote + locator:

a. **Normalized single-cell feature matrix:** Let $X \in \mathbb{R}^{n \times p}$ be a dataset

with n cells and p highly variable genes, where X_{ij} denotes the denoised library-size-adjusted gene expression of gene j in cell i .

Broadly speaking, we use Scanpy [59] to determine the highly variable genes, and use scVI [60] or peakVI [61] to normalize the count matrix of RNA or ATAC, respectively, whenever possible. If the count matrix is unavailable, a log-normalized matrix can be used as input.

Notes:

Reviewer initials:

Review completed:

Article and approved evidence links	Reviewer 2 independent record
HV-030 FORMAL VALIDATION Epigenetic biomarkers in inflammatory bowel diseases—computational challenges and opportunities 2026 Journal of Crohn's and Colitis DOI: 10.1093/ecco-jcc/jjag022 Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: No Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: A range of methods now exists to harmonize datasets before biomarker discovery. Mutual Nearest Neighbours and Harmony, originally developed for single-cell integration, align samples or cells across batches by finding those with similar molecular profiles and adjusting embeddings accordingly.^{128,129} Probabilistic modeling approaches such as scVI and more classical tools like ComBat explicitly model batch-specific technical factors alongside true biological variability.^{130,131} Lessons from single-cell atlas studies emphasize the importance of careful quality control and integration if combined datasets are to be used for downstream biomarker work.¹³² Notes: Reviewer initials: Review completed:

HV-011 FORMAL VALIDATION JIND-Multi: Leveraging Multiple Labeled Datasets for Automated Annotation of Single-Cell RNA and ATAC Data 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.01.15.633130 Article page Full text 1	Review status: NOT STARTED Citation validity: yes Method executed: No Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Various integration methods have been proven effective, such as Harmony [3] and scVI [4] for scRNA-Seq, while Seurat [5] and LIGER [6] can be used for both scRNA-Seq and single-cell ATAC sequencing (scATAC-Seq) data. Notes: Reviewer initials: Review completed:
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HV-050 | FORMAL VALIDATION

Batch correcting single-cell spatial transcriptomics count data with Crescendo improves visualization and detection of spatial gene patterns

2025 | Genome biology

DOI: 10.1186/s13059-025-03479-9

[Article page](#)

[Full text 1](#)

Review status: NOT STARTED

Citation validity: Yes

Method executed: Yes

Use labels (select all that apply):

Primary use: Benchmark

Biological insight: No

Evidence sufficiency: Yes

Supporting quote + locator:

We next compared Crescendo's ability to batch correct individual genes in this mouse brain spatial dataset to five representative state-of-the-art algorithms: ComBat-Seq, scVI, Seurat anchor integration, MNN, and limma (" [Methods](#)"). With the caveat that these methods were not explicitly intended to be used for this purpose, we observed that scVI, Seurat, MNN, and limma struggled to remove batch variation from gene expression (sometimes even increasing of batch variation) or resulted in a dramatic decrease in biological cell-type variation (Fig. [2J](#)).

Notes:

Reviewer initials:

Review completed:

Article and approved evidence links	Reviewer 2 independent record
HV-004 FORMAL VALIDATION Hybrid deep learning–mechanistic modeling of cellular dynamics from a spatiotemporal single-cell atlas 2025 DOI: 10.1101/2025.10.02.679951 Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: No Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Deep generative models such as single-cell variational inference (scVI) (13) and continuous-time neural model (2) motivate our “VAE with latent neural ODE” design. Notes: Reviewer initials: Review completed:
HV-003 FORMAL VALIDATION BrainBridge Characterizes Key Factors affecting Alzheimer’s Disease and Associated Phenotypes 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2025.12.03.692255 Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: No Use labels (select all that apply): Primary use: Extend or develop Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Furthermore, we also include MrVI [25] as a new baseline method. MrVI is a computational tool developed based on scVI [39] to model sample-level covariates. Notes: Reviewer initials: Review completed:

HV-019 | FORMAL VALIDATION

Anti-bias training for (sc)RNA-seq: experimental and computational approaches to improve precision

2021 | Briefings in Bioinformatics

DOI: 10.1093/bib/bbab148

[Article page](#)

[Full text 1](#)

Review status: NOT STARTED

Citation validity: Yes

Method executed: No

Use labels (select all that apply):

Primary use: Mention only

Biological insight: No

Evidence sufficiency: Yes

Supporting quote + locator:

For instance, dimension reduction based on zero-inflated distributions was used on scRNA-seq datasets to extract informative variables for further analysis (e.g. ZIFA [[105](#)], ZINB-WaVE [[106](#)] or ScVI [[107](#)]).

Notes:

Reviewer initials:

Review completed:

Article and approved evidence links	Reviewer 2 independent record
HV-005 FORMAL VALIDATION Single-cell integration and multi-modal profiling reveals phenotypes and spatial organization of neutrophils in colorectal cancer. 2026 DOI: 10.1016/j.ccell.2025.12.003 Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: Yes Use labels (select all that apply): Primary use: Apply to Biology data Biological insight: Yes Evidence sufficiency: Yes Supporting quote + locator: . Integration of samples was done on the filtered counts matrix with an scvi model of one hidden layer and 30 latent components. Notes: Reviewer initials: Review completed:
HV-017 FORMAL VALIDATION Multi-omic identification of perineurial hyperplasia and lipid-associated nerve macrophages in human polyneuropathies 2025 Nature Communications DOI: 10.1038/s41467-025-62964-8 Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: Yes Use labels (select all that apply): Primary use: Apply to Biology Data Biological insight: Yes Evidence sufficiency: Yes Supporting quote + locator: Next, we used atomic sketch integration (method LeverageScore, 5000 cells) in Seurat to reduce memory usage. Batch effects were accounted for by integrating the samples with scVI v1.0.4⁵⁴. After inspecting the effect of batch removal, the full dataset was integrated with Seurat. The scVI integrated full dataset was used to calculate the UMAP embeddings with Seurat (30 dimensions). We identified clusters with the FindNeighbors and FindClusters (resolution 0.7) functions. Notes: Reviewer initials: Review completed:

HV-035 | FORMAL VALIDATION

Sequencing dropout-and-batch effect normalization for single-cell mRNA profiles: a survey and comparative analysis

2020 | Briefings in Bioinformatics

DOI: 10.1093/bib/bbaa248

[Article page](#)

[Full text 1](#)

Review status: NOT STARTED

Citation validity: Yes

Method executed: Yes

Use labels (select all that apply):

Primary use: Benchmark

Biological insight: No

Evidence sufficiency: Yes

Supporting quote + locator:

Demonstrated by Figure 9, for the datasets containing two batches and one group, Batchelor and Beer are able to moderate and reduce the batch effects on both of the medium and high batch effect situation. The low Silhouette score calculated using the batch index and cell embedding in tSNE plot also indicates a high overall similarity between these batches after normalization. ScVI and Seurat can reduce the batch effect to a great extent on the datasets simulated with a medium degree of batch effect. However, on the datasets with high degrees of batch effect, the high Silhouette scores for scVI and Seurat suggest that relatively large differences still remain between the batches even after normalization.

Notes:

Reviewer initials:

Review completed:

Article and approved evidence links	Reviewer 2 independent record
HV-044 FORMAL VALIDATION Toward universal cell embeddings: integrating single-cell RNA-seq datasets across species with SATURN 2024 Nature Methods DOI: 10.1038/s41592-024-02191-z Article page Full text 1 Full text 2	Review status: NOT STARTED Citation validity: Yes Method executed: Yes Use labels (select all that apply): Primary use: Benchmark Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: error bars represent standard error. For batch correction methods (Harmony, scVI and Scanorama), the input genes are selected as the one to one homologs determined by ENSEMBL. Notes: Reviewer initials: Review completed:
HV-031 FORMAL VALIDATION Candida albicans exhibits distinct cytoprotective responses to anti-fungal drugs that facilitate the evolution of drug resistance 2020 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2020.01.21.914549 Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: Yes Use labels (select all that apply): Primary use: Apply to Biology Data Biological insight: Yes Evidence sufficiency: Yes Supporting quote + locator: Normalization, imputation and batch correction were performed by scVI^{58,59}, a tool which implements a probabilistic model of mRNA capture and uses a variational autoencoder to estimate priors across batches and conditions (Supplemental Figure 1). Notes: Reviewer initials: Review completed:

HV-032 FORMAL VALIDATION Effectively Clustering Single Cell RNA Sequencing Data by Sparse Representation 2021 IEEE Transactions on Computational Biology and Bioinformatics DOI: 10.1109/tcbb.2021.3128576 Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: No Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Lopez <i>et al.</i> [20] introduced a single-cell variational inference (scVI) to derive the probabilistic representations of scRNA-seq data by deep generative model variational auto-encoder. Notes: Reviewer initials: Review completed:
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Article and approved evidence links	Reviewer 2 independent record
HV-045 FORMAL VALIDATION An overview of computational methods in single-cell transcriptomic cell type annotation 2025 Briefings in Bioinformatics DOI: 10.1093/bib/bbaf207 Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: NO Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: This strategy was later adopted by deep learning methods, such as single-cell variational inference (scVI) [100], where the encoder preferentially processed the HVG subset. Notes: Reviewer initials: Review completed:
HV-048 FORMAL VALIDATION Beyond the List: Bioagent-Agnostic Signatures Could Enable a More Flexible and Resilient Biodefense Posture Than an Approach Based on Priority Agent Lists Alone 2021 Pathogens DOI: 10.3390/pathogens10111497 Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: No Use labels (select all that apply): Primary use: Mention only Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: Scientists also need better models to eliminate nuisance technical variation that arises between different runs of the same single-modality 'omics experiment (i.e., batch effects) [107,108]. Notes: Reviewer initials: Review completed:

HV-021 FORMAL VALIDATION Biologically Informed Variational Inference Enables Interpretable Cell Phenotyping and Discovery 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.06.10.657924 Article page Full text 1	Review status: NOT STARTED Citation validity: Yes Method executed: No Use labels (select all that apply): Primary use: Extend or Develop Biological insight: No Evidence sufficiency: Yes Supporting quote + locator: NetworkVI generates a rich, batch-corrected low-dimensional representation of uni-, bi-, and trimodal single-cell datasets, utilizing normalized input data to estimate this representation (Figure 1a). Pairs of decoders and encoders power the integration of each modality. NetworkVI builds upon the multiVI model (Ashuach et al. 2023) and thus leverages modality-specific decoders; scVI for transcriptome (Lopez et al. 2018), PeakVI for chromatin accessibility (Ashuach et al. 2022), and totalVI for epitope data (Gayoso et al. 2021). Instead of fully-connected encoders, NetworkVI employs a prior knowledge-driven neural network architecture within the encoder. Notes: Reviewer initials: Review completed:
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Article and approved evidence links	Reviewer 2 independent record
HV-046 FORMAL VALIDATION Projecting single-cell transcriptomics data onto a reference T cell atlas to interpret immune responses 2020 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2020.06.23.166546 Article page Full text 1	Review status: NOT STARTED Citation validity: No Method executed: No Use labels (select all that apply): Primary use: Not sure Biological insight: No Evidence sufficiency: No Supporting quote + locator: Notes: Reviewer initials: Review completed:

HV-024 | FORMAL VALIDATION

A multi-dataset single-cell meta-analysis of human keloid skin across Asian, Black and White populations reveals endothelial and mesenchymal programs linked to ethnic disparities

2026 | Frontiers in Medicine

DOI: 10.3389/fmed.2026.1751725

[Article page](#)

[Full text 1](#)

Review status: NOT STARTED

Citation validity: Yes

Method executed: Yes

Use labels (select all that apply):

Primary use: Apply to Biology Data

Biological insight: Yes

Evidence sufficiency: Yes

Supporting quote + locator:

Despite technical heterogeneity, scVI integration successfully harmonized datasets while preserving biological variation. UMAP visualization revealed clear separation of major skin compartments: keratinocytes (27.3%), fibroblasts (24.8%), endothelial cells (11.2%), *T* cells (9.7%), myeloid cells (8.3%), and other populations (**Figures 1D,E**). Integration metrics confirmed successful batch correction (average silhouette width: condition = 0.31, study = 0.02; kBET rejection rate < 0.05) while maintaining biological signal (cell type LISI = 1.02).

Notes:

Reviewer initials:

Review completed: